BIOLOGY

QUESTIONS AND SUGGESTED SOLUTIONS

COMPILED BY
TEAM LEGENDS

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TRUST THE PROCESS

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PART ONE

MICROSCOPE

1. **Define the following as used in microscopy**
   a. **Magnification**
      - Magnification refers to the enlargement of an object by the objective and ocular lens.
   b. **Resolution**
      - This refers to ability of the microscope to distinguish two objects which are a distant apart.
   c. **Contrast**
      - It refers to the microscope's ability to distinguish an object from its background.

2. **How does a fluorescence microscope produce images?**

- Fluorescence microscope uses the ability of an object to observe wavelength of light at a shorter distance and give off light waves at a longer distance.
- Filters transmit light emitted by fluorescently stained molecules and
- It produces a bright image against a yellow background.

3. **Which microscope can be used in viewing Taenia saginata.**
   - Light microscope (dark field microscope)
4. **After observing the cell, you would like to determine the temperature of the cell. Which microscope would you use?**
   - Scanning probe microscope
5. **Which microscope can be used in Nano level observations as well as cell morphology?**
   - Transmission electron microscope
6. **What are some of the ways of increasing contrast in optic microscopes?**
   - By staining
   - By phase contrast illumination
   - By bright field illumination
   - By dark field illumination.
7. **What is working distance?**

- This refers to the distance between the front lens of the objective and the object on the slide.

8. Which objective(s) provides the least magnification?

- $\times 10\ and\ \times 40$

9. Define the following terms.

a. Refractive index

- This is the measure of light bending ability as it travels from one media to another e.g. water to air.

b. Resolution.

- This is the ability of the microscope to distinguish two objects which are a distant apart.

c. Numerical aperture

- This is the mathematical expression that explains how the condenser lens concentrate the light rays into the objective lens.

10. Differentiate simple and compound microscope and explain how each make images.

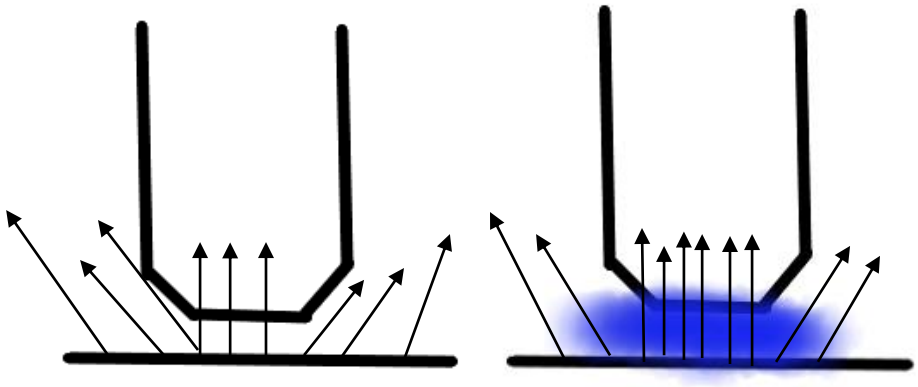
- A simple microscope consists produces an image through a single magnification lens while A compound microscope produces an image through a series of magnification lens.

- A simple microscope has a highest magnification of × 40 while A compound microscope has a highest magnification of × 100

Image production

- A simple microscope produces an images when light rays pass from the specimen through the objective lens to the ocular lens through a single tube which consists of objective lens and ocular lens.
- A compound microscope produces an image when light rays are concentrated in the objective lens by the condenser which is below the stage. Then, the objective lens transfers the light rays to the ocular lens, hence image production.

11. With the aid of a well labelled diagram, describe how an immersion oil in optical microscope is used to achieve higher magnification with good resolution.



- Oil immersion shortens the working distance hence allowing more light rays to enter the objective lens thereby improving resolution of the microscope.
- The immersion oil also reduces the loss of light rays since oil immersion has a low diffraction rate of light rays.

12. How does a scanning microscope differ from a transmission microscope?

- A scanning electron microscope has low magnification while a transmission electron microscope has a high magnification of up to Nano level
- A scanning microscope produce 3 dimensional images while a Transmission microscope produces 2 dimensional images
- Due to its high resolving power, a transmission microscope can show cell morphology unlike scanning microscope.

13. Which microscope technique relies on the specimen interfering with the wavelength of light to produce a high contrast image without the need of dyes or any damage to the sample.

- Fluorescence microscope

14. The field of view of a $\times$ 10 ocular and $\times$ 4 objective is 5mm, what will be the field of view with a $\times$ 10 objective.

Solution:

If $\times 4\ objective = 5mm$

$\times 10\ objective = less$

$= \frac{\times 4}{\times 10} \times 5mm$

= 2mm

(the answer is correct because the higher (greater) the objective, the smaller the field of view; and vice versa)

15. If the ocular lens of a microscope is $\times$ 10 and the objective is set at $\times$ 100, what is the total magnification of the microscope.

$magnification = ocular\ lens\ \times objective\ lens$

M = ($\times$ 10) $\times$ ($\times$ 100)

M = $\times$1000

16. State the function of the condenser.

- It concentrates light rays coming from the light source, into the objective lens.

17. The lens that is within the eyepiece is called…

- Ocular lens

18. The wheel under the stage that adjusts the amount of light is called…

- A diaphragm

19. The high power objective works with which knob?

- Fine focus knob

20. The type of microcopy technique which gives detailed examination of internal structures apart from electron microscope is

- Phase contrast microscopy technique

21. Give one advantage and one disadvantage of the following microscope techniques.

a. Electron microscope

- Advantage: electrons have shorter wavelength, hence higher resolving power.
- Disadvantage: cannot to be used to view live cells as the specimen is placed in a vacuum, hence the cell can be killed.

b. Light/ optical microscope

- Advantage: requires minimal preparation
- Disadvantage: limits diffraction to $0.2\mu m$

c. Fluorescence microscope

- Advantage: has high resolving power since it uses ultra-violet rays.
- Disadvantage: dyes (flourochromes) may kill the cell.

22. What do phase contrast and dark field microscopes have in common?

- No staining is required; hence it can be used for viewing live cells.

23. What are some of the drawbacks of electron microscope?

- It is not user friendly, i.e. it requires maximum specimen preparation.
- It cannot be used to view live cells.

24. Which objective provides the greatest depth of view?

- ×10 objective lens.

25. Three limitations of bright field microscope apart from poor contrast are:

- Limits diffraction to $0.2\mu m$
- Cannot be used to view live cells.
- Stains used, can kill the specimen.

26. How does oil immersion help in achieving good resolution?

- Oil immersion shortens the distance between the slide and front lens of the objective, hence reduces the number of light rays which are lost

due to refraction. This helps lighter rays to be go to the objective lens leading to higher resolution.

27. Briefly explain why the knowledge on the movement of light through convex lens is important in microscope.

- This knowledge helps a person to understand and to use necessary lenses in order for magnification to take place. A convex lens magnifies at a closer range than a human eye hence providing magnification. It does this by collecting the light rays and converging them at a focal point, for magnification.

28. State four factors that help in improving resolution in light microscope.

- Using immersion oil.
- Using short wavelengths of light.
- Increasing the numerical aperture.
- Keeping the diaphragm opened at its best position.
- Keeping the condenser at the highest level, to prevent loss of light rays due to refraction

29. Describe the advantage of using a stereoscopic or dissecting microscope.

- Provides a large field of view and large depth of view as its maximum magnification lens is ×40.
- The specimen may be manipulated (turned, flipped) while under view.

30. Differentiate simple from differential staining techniques in microscope.

- Simple staining uses a single dye which is attainable by all organisms in a smear while differential staining techniques uses the ability of the organisms to interact with different dyes.

31. How are microscopes classified?

- Based on ocular lens (mono-ocular & binocular lens)
- Based on number of lenses (simple or compound)
- Depending on source (light & electron)

32. Differentiate the two types of specimen fixation.

- Heat fixation: where an organism is placed on a slide and it (the slide) is heated from a heat source.
- Chemical fixation: it where an object is placed on a slide and a chemical e.g. alcohol or ethanol

is applied in order to deactivate chemical processes happening on the slide.

33. Differentiate between optical and electron microscope.

- Optical microscope has lower resolving power than electron microscope which can magnify up to $1 \times 10^6\ times$.
- Optical microscope can be used to view live cells whilst an electron microscope can kill the specimen, hence cannot be used to view live cells.

34. What are the three branches of microscope?

- Light microscope
- Electron microscope
- Scanning probe microscope

35. Who invented the compound microscope?

- Robert Hooke

36. You want to reduce the amount of light when looking at a thin and\or transparent cells or tissues such as onions or amoeba cells. What options would you use to decrease the light intensity?

- Closing the diaphragm, a little bit.

37. Define the term parfocal.

- This means that an object can still be focused while changing from one objective lens to another.

38. Outline the two differences between compound microscope and transmission electron microscope.

- Compound microscope can be used in viewing live cells while a transmission electron microscope cannot be used to view live samples.
- Compound microscope comprises of a number of convex lenses which can magnify a whole object leading to a better view while a transmission microscope has a thin series of convex lens hence producing 2D images.

39. Define field of view

- This refers to the diameter of the objective lens under use when viewing a sample.

40. Leuwenhoek was able to have improved microscopic observations than Hooke's observations; why?

- Leuwenhoek used live cells than Hooke's observations.

41. Differentiate wet mounts and dry fixation as techniques used in microscopy.

- Wet mount is a type of fixation, where a drop of liquid (water) is dropped on the specimen on the slide. While dry fixation, a specimen is put on a slide and no any liquid is applied.

42. Cilia and flagella motility are some of the features which can enable pathogenic prokaryote cell to attach it to a host cell more easily. Given a situation that you want to observe the flagella and cilia, which best microscope would you use.

- Phase contrast microscope. Due to its ability to view live cells and cell's organelle motility, it can be used to view the cilia and flagella. It does not use stains for magnification hence providing a good view for cilia and flagella's motility.

43. **Photographs of cells of any particle taken with a microscope are called** microscopic particles

44. **Magnification in microscope is affected by which two lenses.**

- Eyepiece lens and objective lens

45. Briefly, discuss about the three advantages of light microscope

- It is cheap
- It requires minimal sample preparation
- It uses white light for sample preparation.

- It is user friendly
- It is parfocal

46. How does a phase contrast microscope produce an image?

- A phase contrast microscope uses the variations in density within the cell.
- It uses the ability of interference of light rays coming out from the sample. The more the refractive index of a cell organelle, the poor the resolution.
- Light rays coming from the sample diverge and meet at the focal point forming an upright image.

47. How does a bright field microscope produce an image?

- A bright field microscope forms a dark image against a bright background.
- Light rays coming from the light source are converged on the sample and the sample refracts the light rays in the objective lens and to ocular lens, hence forming an image.

48. What is the use of a scanning probe microscope?

- Scanning probe is used in viewing details of molecules e.g. DNA, RNA or viruses.

49. How does an electron microscope produce an image?

- When electrons are fired from the electron gun, the protons which are found alongside the tube in the microscope concentrate and focus the electrons in the first condenser.
- The electrons past through the first condenser lens to the second condenser lens.
- The electrons then pass through the specimen and form an image on the transparent screen.

50. What happens on phase contrast illumination?

- A sample's contrast is improved when there is light interference coming from different parts of the specimen.

PART TWO

CELL BIOLOGY

51.Briefly discuss three differences between prokaryotic and eukaryotic cells?

- According to their genetic structure, in prokaryotic cells mitotic nuclear division is absent while in eukaryotic cells is present.

- DNA in organelle is absent inn prokaryotic cells while in eukaryotic cells is present.

- In their cytoplasmic structure prokaryotic cells have 70s nature of cytoplasmic ribosomes while eukaryotic cells have 80s cytoplasmic ribosomes.

52.State one cell organelle for each of the following cellular functions.

a) **Lipid synthesis** => smooth endoplasmic reticulum.
b) **Proteins synthesis** => ribosomes.
c) **Energy ATP production**=> mitochondria.
d) **Separation of daughter cells after cell division**=> spindle fibers.

e) **Regulation of calcium levels**=> endoplasmic reticulum.
f) **Digestion of worn out organelles engulfed viruses and bacteria**=> lysosomes.

53.Using well labelled diagram discuss the process of conjugation.

- In conjugation two paramecium cells meet and form a bridge and one the cells generally donate some replicated DNA. this allows sharing of genes among paramecium even among different genera.

54.How does cytokinesis in animal cell differ from cytokinesis in plant cells?

- In animals, membranes pinch together to form a cleavage furrow while in plants, vesicles containing cellulose form and fuse between two daughter cells thereby forming a complete cell wall.

55.Peroxisomes are often called "suicide bags" because of their ability to destroy the cell. How do cells contain the peroxisomes without being destroyed?

- Peroxisomes oversee reactions that neutralize free radicals. They produce large amounts of the toxic H_2O_2 in the process, but contain enzymes that convert H_2O_2 into water and oxygen. These by products are then safely

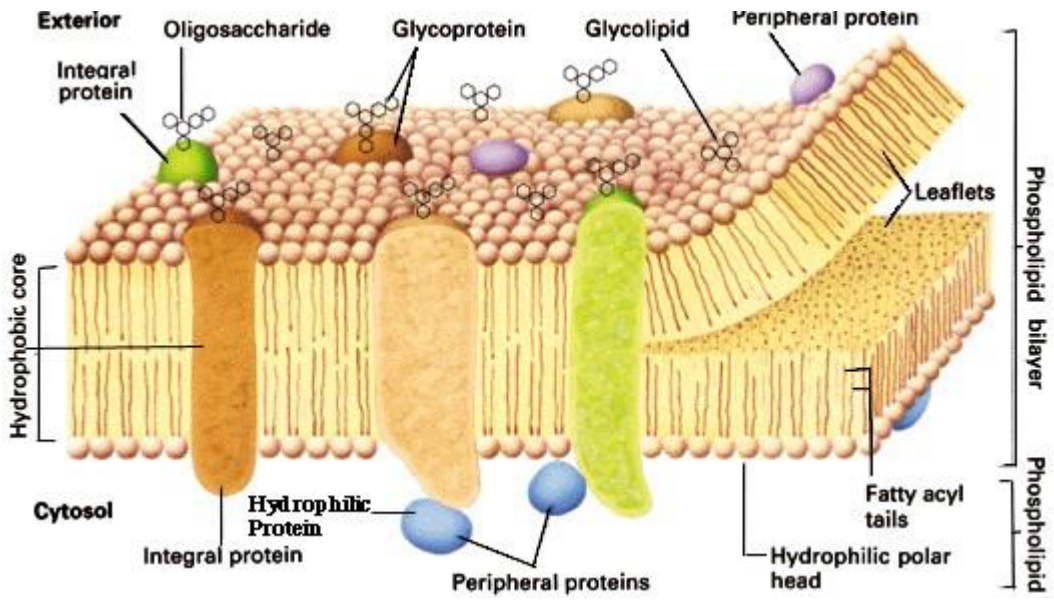
released into cytoplasm. Therefore, peroxisome neutralize harmful toxic so that they do not cause damage in the cell.

56.In which two ways is meiosis related to sexual reproduction?

- Meiosis produce sex cells that joins to form new offspring's during sexual reproduction.

- Meiosis allows offspring's produced during sexual reproduction to grow and develop.

57.All cells have cell membranes and other structural components. Using a well labelled diagram, describe the structure of the cell membrane?



- It is composed of bilayer of phospholipid molecules and variety of membrane proteins. They have a glycocalyx

consisting of oligosaccharide molecules projecting from the glycolipids and glycoproteins.

58.Explain one major role played by the cells membrane in maintaining homeostasis?

- Cell membranes prevents the passage of water and ions. This allows the cell to maintain a higher concentration of sodium ions out the outside of the cell.

59.Differentiate the following in terms of their function or general structure?

a)Protoplasm and cytoplasm

- cytoplasm involves everything to the inside of the cell membrane the nucleus while protoplasm involves cytoplasm plus nucleus.

b)Smooth and rough endoplasmic reticulum.

- Smooth endoplasmic reticulum is prominent in steroid secreting cell such as liver, active mammary glands and intestinal cells. While rough endoplasmic reticulum is prominent in cells that make large amount of proteins to be exported from the cell or inserted to the cell membrane and cells circulating enzyme such as pancreas.

60.Discuss the typical structure of bacteria cell?

- They are unicellular microscopic prokaryotes size range between 0.5-2µm, have mesosomes for photosynthesis and

respiration sites, they also have small rings of DNA called plasmids, its cell wall is made up of peptidoglycan and have flagella for movement.

61.List three major differences between bacteria cell and plant cell?

- Plant cell is eukaryotic while bacteria are prokaryotic
- DNA in Plant cell is arranged in nucleus in contrast to bacteria DNA found in cytoplasm
- Plant cell contains membrane bound organelle whereas bacteria cell lacks membrane bound organelle

62.Give one reason why daughter cells produced by mitosis are genetically identical unlike cells produced by meiosis?

- It is because crossing over has occurred; the events that occur in meiosis but not in mitosis include homologous chromosomes pairing up, lining up along the metaphase plate in tetrads.

13.Define the following terminologies?

a) Peroxisomes

- Are membranes bound organelles containing enzymes that catalyze an assortments of metabolic reactants in which hydrogen is transferred from various compounds.

b) Centrioles

- Are small structures made of microtubes that exist as part of the centrosome which helps organize micro tubes in the body.

c) Chromatin

- Are complex proteins associated with DNA which appears as network of granules and strands in cells that are not dividing

14.Explain the general events that take place during each of the following stages of the cell production?

S1. NO	STAGES	EVENTS
	CELL CEYCLE (3marks)	
1	G1	Growth and normal metabolic processes
2	S	DNA replication
3	G2	Growth and preparation for mitosis
	MITOSIS (14marks)	
4	Prophase	- Chromosomes are shorten and become visible in animals. - Centrioles move to opposite sides of the cell in animals

		- Spindle fibers and astral fibers both known as the spindle apparatus begin to form
5	Metaphase	- Chromosome line up along Centre of the cell called the metaphase plate in animals - Chromosome attach to spindle fibers in animals. - Spindle and astral fibers are now clearly visible in plants.
6	telophase	- Nuclear envelop forms around both sets of chromosomes in animals - DNA uncoils in animals - Spindle and astral fibers completely disappear - Cytokinesis happens
	Meiosis (3marks)	
7	Meiosis1	- There is chromosomal replication in interphase, - Nuclear membrane disappears an d spindle fibers appear

65.Briefly describe the cell theory and relate it to evolution?

- According to cell theory it states that every cell comes from another cell, so due to changes in the environment the

cells might have undergone some changes in genes chromosomes which lead to variations among them, hence making the cells to change gradually over a long period of time as the genes pass on to another generation.

66.What are the three central dogma of cell theory?

- It state that all living things are composed of one or more cells
- Cells come only from already existing cells
- Cells are the basic units of structure and function in an organism

67. Which cellular feature would determine whether a cell is prokaryotic or eukaryotic?

- Presence of cell membrane

68.Use the figure below to answer the questions that follow:

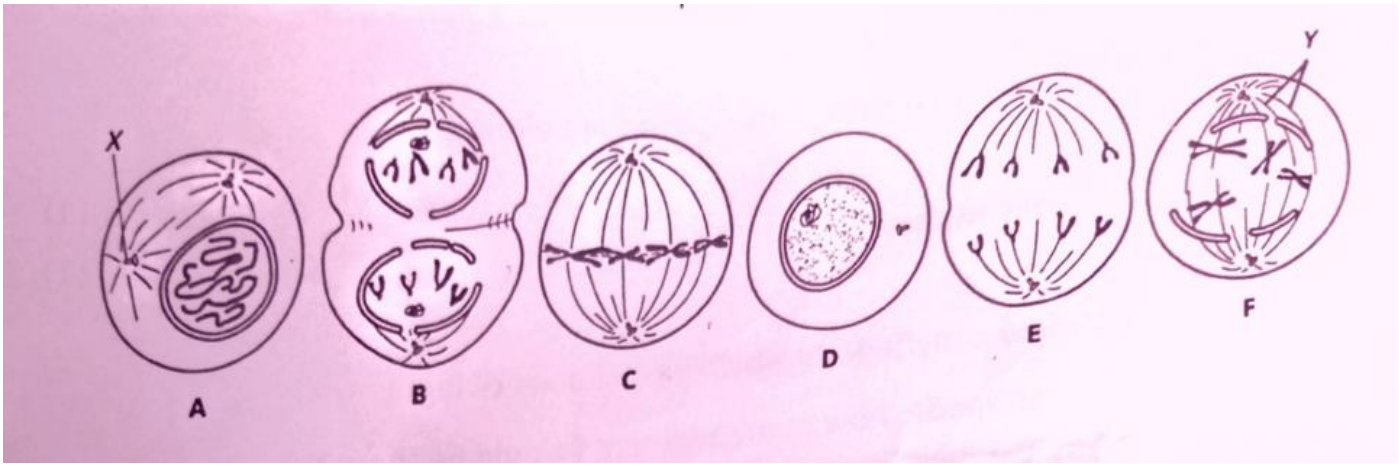


Figure shows a cell undergoing mitosis

a) **What phase of mitosis is represented by cell A?**
   - Metaphase

b) **What phase of mitosis is represented by cell B?**
   - Telophase

c) **What type of cell organelle form part y in cell F?**
   - Centrioles

d) **In cell F, what is the name of structure labelled y?**
   - Spindle fibers

69. In 1610, Galileo Galileo invented the microscope, but it was Robert Hooke in 1665 who was the first to observe cells why?

- It is because Robert Hooke improved the microscope by developing a concave and convex lens so as to magnify small organisms such as cells.

70. No one has disputed the cell theory but what do you think are some of the limitations of the cell theory.

- It fails to explain the variations that exist among different organisms (parents and offspring's).

- It fails to describe the origin of the first cell.

71. Why was Archaea considered as a Prokaryote?

- It does have an enclosed nucleus

- It does not have a membrane bound organelles.

72. How is Archaea different from Bacteria?

- Archaea has glycerol while Bacteria has peptidoglycan.

73. Briefly discuss the different types of transport in cells.

- Active transport involves the use of energy to transport particles against concentration gradient.

- Passive transport does not involve the use of energy to transport the particles and molecules from high to low concentrated region. e.g. Diffusion and Osmosis.

74. Discuss with examples two ways by which the knowledge about the cells is applied to daily

life.

- Cells can be used to replicate the viruses for production of drugs and medicine e.g. vaccines.

- It can be used for generation of clean burning fuels as a renewable source of energy such as Methane and Hydrogen.

75.Cell reproduce through a continuous sequence of growth and division called the cell cycle.

a) **Mention the four stages of a cell cycle.**
   - First growth phase
   - Synthesis phase
   - Second growth phase
   - Mitotic phase

b) **During which stage does DNA replication occur?**
   - Synthesis phase

76.Phases of meiosis in living cells are generally similar to those of mitosis. However, meiosis involves two sequences of phases (meiosis I and meiosis II). Answer the following questions in relation to the processes mitosis and meiosis;

a) **What is centromere?**
   - It is a point of attachment between two chromatids

b) Describe the difference between metaphase I and metaphase II.

- In metaphase I the replicated homologous chromosomes are aligned at the middle of the cell while metaphase II the sister chromatids are brought on the metaphase plate by the action of microtubule.

c) Explain what happens to chromatids for chromosomes to appear during cellular reproduction.

- They become thick and short.

77.State two cell structures that are characteristic for all cells.

- Both have nucleus which contain genetic material
- Both have cell membrane

78.What cell organelle contains small amounts of DNA?

- Plasmids

79.Relate the structure of the cell membrane to its function.

- The structure of a cell membrane is composed lipids and protein layers; hence its function is semi-permeability

and these lipids and proteins regulate the movement of particles across the cell membrane.

80.Explain why two species can have the same chromosome number yet have very different attributes.

- During the cross over, difference characteristics carried by different chromatids may be transferred to the other chromatids hence changing the attribute of the final cell.

81.Distinguish the following terms in relation to cell division:

a) Chromatin and chromatids

- Chromatin is a material consisting of DNA and associated proteins While Chromatids are two strands of DNA that joins at the centromere to form a chromosome

b) Chiasmata and kinetochore

- Chiasmata is a site of crossing over while kinetochore is a point attached to each centromere of a chromosome.

82.What stage of cell division will you expect to find the following?

a) Splitting of centromere into two

- Anaphase II

c) Formation of cell plate

- Telophase I

83.How do animals and plants cells differ in spindle formation?

- Plants cells show a microtubule organizing centers which consist of fibrils with little or no structures while animal cell have a pair of centrioles in the middle of microtubules organizing center and the centrioles are surrounded by fibrils that make up the peri-centriolar material.

84.Explain why is it usually important to conduct the Gram reaction test as an initial test of bacterial identification?

- Gram staining is very accurate and it reduces the artifacts, since you only make conclusion based on the colors which have been observed. If it is crystal violet, then it is gram positive bacteria and when it is colorless, it is gram negative bacteria.

85.You have discovered a new prokaryote and classified it into Archaea kingdom. Briefly describe the major characteristic that might lead you to this conclusion.

- Lack of well-defined nucleus may help in concluding that it is a prokaryote.

86.Tay-sachs disease is a human genetic abnormality that results in cells accumulating and becoming clogged with very large and complex lipids. Which cellular organelle must be involved in this condition?

- Lysosome

87.State one major role played by the following organelles:

a) Ribosomes

- For protein synthesis under the control of the genetic code.

b) Centriole

- Production of spindle fibers that are used in cell division

d) Golgi body

- In plants, it is used for synthesis of cell wall material while in animals is added to nascent proteins coming from the endoplasmic reticulum for use internally for secretion.

e) Cell vacuole

- Contains salts, sugars, pigments and other substances that create osmotic potential in the cell.

f) Mitochondria

- Production of energy in form of ATP through the process of respiration.

88.Distinguish the following:

a) Fertilization and meiosis

- Fertilization is the fusion of an oocyte and a sperm to form a zygote while meiosis is a sexual reproduction which is done to form gametes.

b) Conjugation and binary fission in bacteria

- Conjugation refers to sexual reproduction in which two paramecia meet and exchange genetic materials while binary fission is asexual reproduction in which one cell replicates then cytokinesis happens to form other organisms.

c) Cytokinesis in animal cells and cytokinesis in plant cells

- Cytokinesis in animals the membrane pinch together to form a cleavage furrow. The cleavage furrow fuses and form two daughter cells while cytokinesis in plants cell wall must be formed between two daughter cells so vesicles containing cellulose is formed and fuses between two daughter cells eventually forming a complete cell wall.

d) Rough endoplasmic reticulum and smooth endoplasmic reticulum

- Rough endoplasmic reticulum contains ribosomes which are responsible for protein synthesis while smooth endoplasmic reticulum does not have ribosomes hence it is used for lipid synthesis.

89. Which organelle is considered the "brain" cell?

- Nucleus

90.Which cell organelle assist with cell division during mitosis and meiosis?

- Centrioles

91.When does the disappearance of nucleoli and the nuclear envelope occur during meiosis II?

- During Anaphase II

92.Study the figure below and answer the questions below

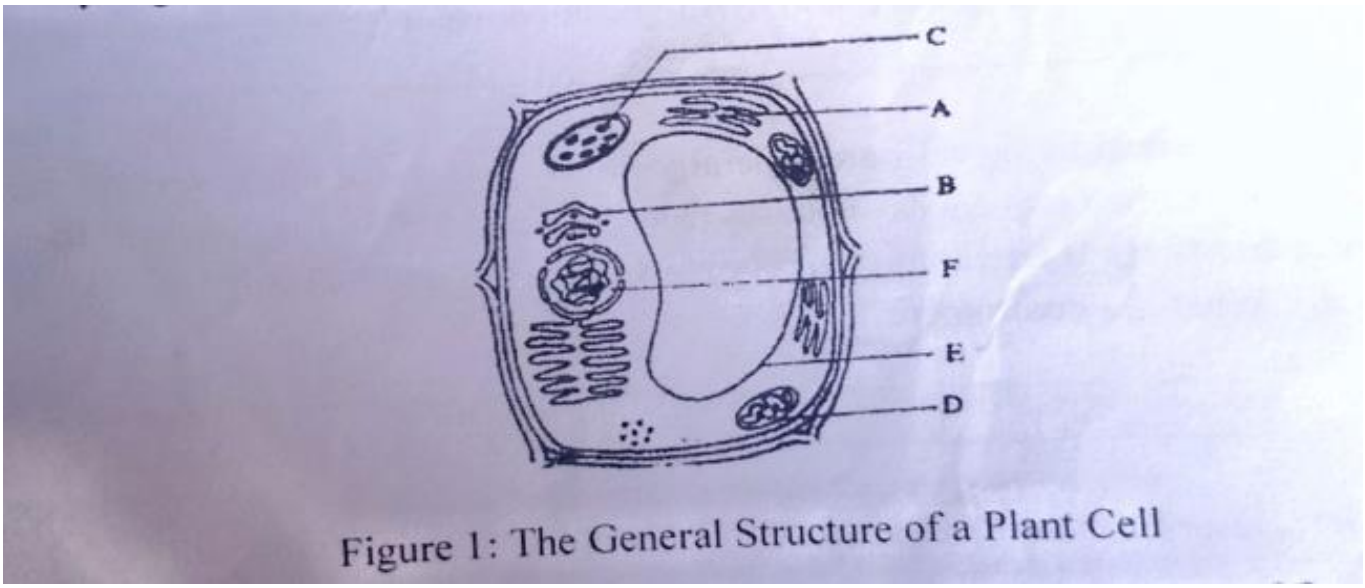


Figure 1: The General Structure of a Plant Cell

a) **Distinguish between part labelled A and B in terms of function**

- Part labelled A is used for synthesis of cell wall material in plants and in animals is added to nascent proteins coming from the endoplasmic reticulum for use internally for secretion while Part labelled B is a site for protein synthesis under the control of the genetic code while.

b) State the function of part labelled C.

- It provides a site for biological and photochemical reactions of photosynthesis.

93.Briefly discuss whether a virus is a living organism or not?

- According to cell theory viruses are living organisms because they show some of the characteristics of living things, but they do not have cells therefore all viruses are acellular organisms.

94. How does a sex cell differ from a body cell?

- A sex cell has half the amount of genetic material as a body cell.

95. How do the chromosomes at the end of meiosis I compare with the chromosomes at the end of meiosis II?

- Chromosomes have two chromatids at the end of meiosis I and one chromatid at the end of meiosis II.

96. How is meiosis related to sexual reproduction?

- Meiosis produces the sex cells that join to form new offspring during sexual reproduction.

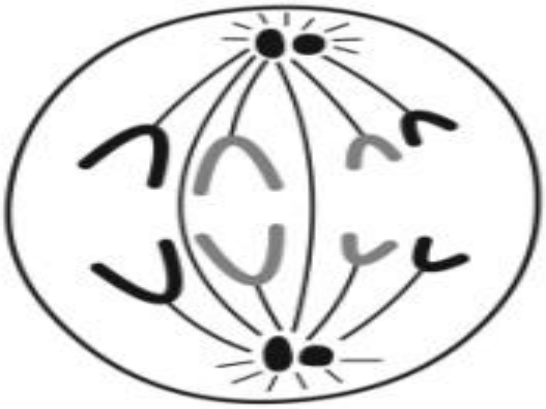
97. How does meiosis I differ from meiosis II?

- Chromosome number decreases by half as a result of meiosis I, but not as a result of meiosis II.

98. What do chromosomes do in meiosis but not in mitosis?

- The homologous chromosomes form pairs.

99. A student observed a slide of a stage of mitosis as shown in the diagram below.



a) **He/she noticed that the slide contains two homologous pairs of chromosomes. How would this diagram be different if it showed anaphase I of meiosis instead of anaphase of mitosis?**

- Each chromosome would still have two chromatids.

b) What is the major significance of mitosis and meiosis?

- Mitosis bring about growth in all organisms while Meiosis facilitates stable sexual reproduction.

100. How would living things differ if cell division took place only for reproduction?

- All living things would be single-celled.

PART THREE

MICROBIOLOGY

101. Distinguish between virion and prion.

- virions refers to the virus particle which contain nucleic acid surrounded by a coat called capsid while prions are infectious proteins.

- virions are generally found within the host cell nucleus an appear to interfere with gene regulation whilst prions can be found on the surface of brain of animals

102.Describe any three features of viruses that are used by various classification systems for grouping animals.

a) **Mode of production**
   - Just like other unicellular organisms' viruses mostly reproduce asexually while other animals it reproduces sexually.

b) **Virulence**
   - Most of the viruses are violent in the sense that when they inject in the host cell they destroy the host cell and colonies the place hence destroys cells completely i.e. HIV in the cd4 cells while other unicellular organisms like bacteria and fungi they are not virulent.

c) **Morphology**

- Viruses have different modes of movement. Some viruses they move by using flagella which is also used in other organisms.

d) Genetic material

- Viruses contains ribonucleic acid (RNA) while in other animals the genetic material is composed of DNA.

103.Distinguish conjugation and binary fusion in bacteria.

- Viruses are used to transfer the DNA (genetic material) of bacterial genes from one bacterial cell into another Normally, a phage contains only its own DNA, however, sometimes a phage incorporates some of the bacterial DNA of its host. Whilst in binary fusion the prokaryote bacteria reproduce asexually, generally by binary fission, a process in which one cell divides into two similar cells.

104. Describe the four classes of fungi in terms of structure and reproduction.

a) Phycomycetes

- Phycomycetes have no septate mycelium.

- Phycomycetes produce asexually by producing sporangium.

- Phycomycetes produce sexually by producing zoospores.

b) Ascomycetes

- Ascomycetes have **septate mycelia.**

- Their reproduction is carried out by both sexual and asexual. Sexual spores (**ascospores),** are contained in a sac called **ascus**. Each ascus carries eight or more ascospores.

- Asexual spores (**conidiospores**) are produced open on various types of reproductive stalks (**reproductive hyphae**). Yeasts (e.g., *Sac-charomyces cerevisiae*, or baker‘s yeast) are classified as ascomycetes. However, yeast form large cells that reproduce asexually by budding or fission, and sexually by conjugation and transformation

c) Basidiomycetes

- Basidiomycetes also have a septate mycelium.

- They produce sexual spores (**basidiospores)** inside a structure called the **basidium**. The basidium could either be hollow (one sac) or subdivided into four sections. e.g., agaricus) belong to the basidiomycete group.

- Asexually, as in ascomycetes, they too reproduce **conidiospores.**

- However sexually they produce spores called basidiospores

c) Deuterostomes

- These produce asexually only by producing conidiospores.

- Have a septate mycelium.

105.Why are archaea considered different from eubacteria and eukaryotes?

- Their cell walls lack peptidoglycan like in eubacteria.

- The lipids in the cell membrane of archaea have different structure than those in all other organisms.

- Archaea have distinctive ribosomal RNA sequence, some of their genes possess intrins, unlike of other bacteria.

106.State any seven roles of microbes in mankind.

- Food spoilers

- Industrial use

- Digestion process

- Soil enrichment

- Detoxification of substance
- Used in the production of medicine
- Provide oxygen when doing photosynthesis.

107. List down common shapes of true bacteria

- Cocci
- Streptococcus
- Vibrio
- Spirillum

108. List any two types of bacteria.

- Cyanobacteria
- Sheathed bacteria
- Stalked bacteria
- Budding bacteria

109. How do virus reproduce?

- Virus reproduce asexually through binary fission. They replicate when they are in host cells.

110. Describe how lytic viral cycle from lysogenic viral cycle.

a) In a lytic cycle,

- The virus lyses (destroys) the host cell after integrating with the host cell's genome.

- Viruses that have only a lytic cycle are described as virulent which means that they cause disease, and often cause death on the other hand.

b)In lysogenic

- The lysogenic viral cycle involves viruses which are gentle and they may choose to destroy the host cell after integrating with the host cell's genome.

111. What is the difference between monotrichous and peritrichous flagella arrangement?

- Monotrichous flagella is attached only to one side of an organism.
- Peritrichous on the side flagella is covered on the whole body of an organism

112. Describe two significant roles played by peptidoglycan in microorganisms.

- Peptidoglycan makes the cell wall thick through the cross links of the chemicals such as n acetyl-muramic and glutamine
- Facilitation of movement of substances within the cell such as proteins and Lipids

113. Briefly describe basidiomycetes in terms of reproduction.

- Basidiomycetes produce sexually by producing basidiospores.

- They also produce asexually by producing conidiospores.

114. Give three functional differences between a viral particle and a cell.

- A cell controls all processes in a viral while a viral particle controls aa specific function i.e. metabolism

- A cell contains all organelles for an organism to function while viral particle only contains nucleic acid for the virus.

115. List the main four classes of fungi.

- Basidiomycetes
- Phycomycetes
- Ascomycetes
- Deuteromycetes

116. Describe how fungi are classified based on the following:

a) Morphological structures.

- Others use pili for movement whilst others use flagella

b) Mode of reproduction

- Others produce a sexually whilst others produce sexually

I.e. deuterostomes produce asexually

117. State three roles of fungi.

a) Fungi in nature

- Employed in the commercial production of organic acids, vitamin and antibiotics e.g. Penicillin (biochemical product of the fungus Penicillium spp).

- They decompose food, fabrics, leather, dead plants and animals to release oxygen and minerals

- Many fungi form important symbiotic relationships with other organisms such as lichens, mycorrhizae

b) Example of typical virus

- Tobacco mosaic virus
- virus polyhedral
- Human immune virus

118. Discuss the typical stages of viral particle replication using the case of HIV.

a) Retroviruses use reverse transcriptase to transcribe their RNA into DNA; the viral DNA becomes part of the host DNA.

b) When activated, viral DNA uses host enzymes to transcribe viral RNA

c) HIV infects T helper cells, specialized cells of the host's immune system.

i. *The virus attaches to protein receptors, known as CD4, on the plasma membrane of T helper cells.*
ii. *HIV has two identical single-stranded RNA molecules.*

119. Discuss the difference between staphylococcus and streptococcus bacteria flagella arrangement as seen under microscope.

- Staphylococcus bacteria is accumulated like a rump of stones while streptococcus bacteria form a linear pattern.

120. Discuss the typical structure of a bacteria cell.

- Structure of a bacteria cell is rectangular in shape. It also has pili for production. The flagella on the bacteria cell is used for movement.

121. List down the five microorganisms which are studied under a microscope.

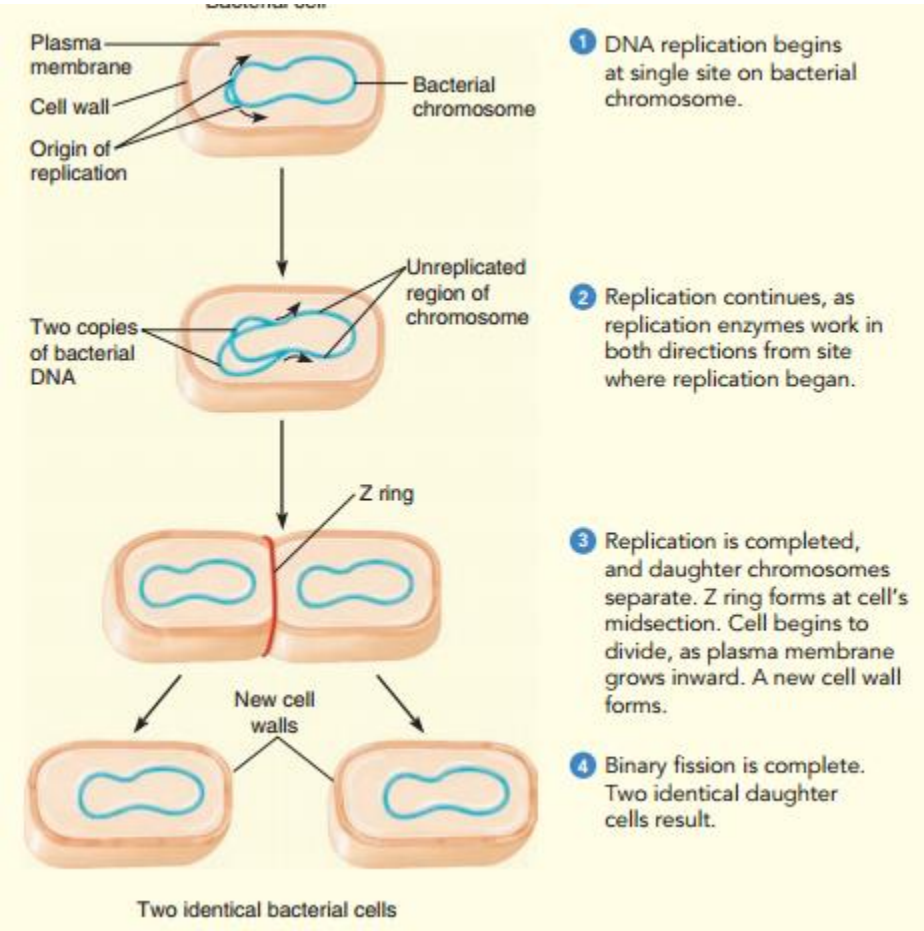
- Virus
- Prions
- Bacteria
- Fungi
- protozoa

122. State the four characteristics that define microorganisms.

- Undefined nucleus

- Contain environmental sensing proteins
- They produce spores when reproducing
- Both are microscopic.

123. Use diagrams to describe the four stages of binary fusion.



124. Explain the three locomotive structures which are used in bacteria.

- paramecium
- The flagella
- cilia

125. Why would you consider basidiomycetes and ascomycetes as sister phyla?

- They both produce conidiophores when producing asexually.
- They both have septate mycelium

PART FOUR

ZOOLOGY

126. Differentiate the following as used in zoology

a. Fertilisation and Implantation

- fertilisation refers to the fusion of a sperm and an oocyte to form a zygote while implantation is the attachment of the zygote in the placenta.

b. **Morula and Gastrula**

- Morula refers to the solid ball of blastomeres while gastrula refers to the process by which the blastula becomes a three-layer embryo.

c. **Ectoderm and Mesoderm**

- Ectoderm is the outermost layer which gives rise to skin and sense organs such as eyes while mesoderm is the middle layer which gives rise to reproductive organs and excretory organs.

d. **Trichinosis and fascioliasis**

- Trichinosis is the disease caused by Trichinella white (white pork) while Fascioliasis is the disease caused by Fasciola (flukes).

e**. Flagella and Cilia**

- Flagella are long but few appendages protruding outside the body of an organism while Cilia are short but many protruding inside or outside the body of an organism e.g. in nose cilia is inside.

f. **Schizogony and conjugation**

- Schizogony is the asexual reproduction of organisms characterised by multiple divisions of the nucleus and cells while Conjugation is a type of sexual production in which takes place when two plamecia join and exchange genetic material.

g. Ovoviviparous and Viviparous

- Ovoviviparous refers to the type of development in which the young hatch from eggs incubated inside the mother's body e.g. sharks while Viviparous is the type of

development in which the young complete their development within the mother's body e.g. all mammals.

- In Ovoviviparous, the young are fully developed when they are hatched and released from the mother while in Viviparous, the young are fully developed inside the mother's body

(Oviparous is the type of development in which the internally fertilized egg is released outside the mother's body to complete their full development e.g. birds.)

h. **Anus and Cloaca**

- Anus is the end and outlet of the digestive tract while Cloaca refers to the common exit chamber from the digestive, reproductive and urinary system (in others, the cloaca may serve as a respiratory duct).

- [a cloaca is a single opening in which feces, urine and reproductive products (gametes) leave the body] *sharks have a cloaca*

I. **Hagfish and Lamprey**

- Hagfish are jawless invertebrates while lamprey are jawless vertebrates.

Hag fish and Lampreys

Tom Stack / Tom Stack & Associates

(a) Three lampreys are attached to a carp by their suction-cup mouths. Note the absence of jaws and paired fins.

▲ **Figure 34.9** A **hagfish**.

127.

a. **Name one phyla of bryophytes**

- Phylum brophyta (mosses)
- Phylum hepatophyta (liverworts)

- Phylum anthocerophyta (hornworts)

b. Name one phyla of gymnosperms

- Phylum genetophyta
- Phylum cycadophyta

128.

a. **Protostomes are characterised by**

- A blastopore which forms the mouth.

b. **Amphibians are characterised by**

- Cutaneous respiration (SKIN RESPIRATION). Most amphibians respire through the skin e.g. frog.

c. **While working in your garden, you discover a worm-like segmented animal with two pairs of jointed legs per segment. The animal is probably a** ……. (an arthropod)

d. **Which phylum is characterised by animals with segmented body?**

- Phylum arthropods

129. What are three traits of gymnosperms which are shared by the angiosperms.

- Both are heterosporous
- Both have companion cells

- Both have sieve plates

130. List two examples of the following:

I. **Asymmetrical animals**

- cnidarian
- echinoderm

II. **Diploblastic animals**

- Cnidarians
- Ctenophores

III. **Acoelomates**

- Platyhelminthes
- Nemertea

IV. **Ascaris**

- Ascaris suum
- Ascaris galli

V. **Hermaphrodites**

- Snails
- Sponges (amoeba)

131. State whether these statements are true or false

Bees are monoecious: **false**

Lungs develop from the endoderm: **true**

A crab is an example of a bilateral symmetry: **true**

A blastocoel is also called the coelom: **false**

Nematode are also known as flatworms: **false**

Loa loa is a nose worm in humans: **false**

Female nematodes are usually larger than the male nematode: **true**

Echinococcosis granulosis forms hydatid cyst in brains: **(false)**

The Heart Develops from The Mesoderm **(true)**

132. What is the difference between deuterostomes and protostomes?

- Protostomes are coelomates whose embryonic development shows a blastopore associated with the mouth. While deuterostomes are coelomates whose embryonic development shows a blastopore associated with the anus.

133. The beef tape worm is one of the parasites of vertebrate animals.

a. The scientific name of the parasite is *TAENIA SOLIUM*

B. The larva stage of the beef tape worm is called

- Cystcercus bovis

c. **Four dangers of beef tapeworms are:**

i. They suck blood leading to anemia

ii. They may cause blindness if they attack eyes

iii. Blocks the intestine hence it can cause death

iv. Cause inflammation if it attacks the tongue.

d. **Humans can control beef tape worms by:**

- Eating food that is well inspected by animal scientists
- Boiling the foods thoroughly.

134. **Name the phylum and class of tapeworms**.

Phylum – platyhelmithes.

Class – cestoda

135. **State the advantages and disadvantages of the exoskeleton in arthropods**.

Advantages

- The exoskeleton protects the organism from external environment.
- the exoskeleton protects the underlying vulnerable parts.

Disadvantages

- The exoskeleton slows the process of molting in an organism.

- It restricts growth of an organism.

136. **Outline the important stages in the life cycle of a plasmodium in the correct order**.

Infection in form of sporozoites ⟶ asexual reproduction to form mesozoites ⟶ sexual production to form gametocytes ⟶ gametocytes form zygotes called ookinites.

137. State the class to which the following belong

a) *Mollusca* – gastropods

b) *Most fish in lake Malawi* – osteichthyes

138.. Suppose you find a myriapoda with 32 pairs of legs

a) If this was a centipede, how many segments would be there?

- (32 ÷2) = 16 segments

b) **Which class do myriapoda belong?**

- Arthropoda

139. How do caecilians and sirens differ from Anura?

- Caecilians and sirens do not have legs while Anura have legs.

- Caecilians have a tail while Anura does not have a tail.

140. Describe adaptation features which enable aves to fly. (5)

- They have feathers. These feathers provide lift in gliding or flight and it also conserves heat.

- They have a lightweight skeleton. This helps the aves in lift.

- They have sufficient respiration. This helps in continual flow of energy to the tissues hence enabling them to fly.

- They are endothermic. Their ability to control internal temperature, enables them to fly in regardless of the external temperatures.

- Presence of the red breast muscles enables the aves to control sustained activity, hence enabling them to fly in any environment.

141. Why do snakes have excellent sense of smell?

-to protect itself from predators

-to hunt prey

142. **What are the three examples of unusual mammals?** And why?

- Marsupials
- Monotremes
- Placenta mammals

143. Name the following

a) **Nematode that is transmitted through ingesting contaminated soil and often causes painful itching around the anus in small children.**

- Round Worms

b) **An adult worm for Cystcercus Bovis?**

- Taenia saginata

144.. Write short notes on the following:

a**. Poriferans**

- These are sponges. Members of phylum porifera. They reproduce asexually by means of binary fission. Amoeba are examples.

b. **Monotremes**

- These are egg laying mammals. Have a well-developed nervous system. They have a four chambered heart.

c. **Urochordata**

- These are marine invertebrates which belong to the phylum of chordata. Jelly fish, star fish are examples.

d. **Chordate characteristics**

- These are deuterostomes with bilateral symmetry and they have three germ layers. A human being is an example of vertebrate.

e. **Amphibians**

- These go through metamorphosis, a transition from lava to adult. They respire through the skin.

145. Explain how animals are classified based on the following:

a. Body symmetry

- Animals are classified based on radial, bi-lateral and asymmetrical body patterns.
- Radial body pattern is where multiple plane is drawn on an animal where mirrors are produced include jelly fish.
- Bi-lateral pattern is where a single plane is drawn on an animal and two equal left and right parts are produced e.g. a human being.
- Asymmetrical pattern is where a body grows without a regular pattern.

b. Body cavity

- Animals are classified based on the coelom present i.e. acoelomates, pseudocoelomates and coelomates

c. Notochord

- Some animals have a longitudinal rod; firm and flexible, but it supports the body. E.g. sharks
- Some animals don't have the skeletal rod e.g. dogs.

146. State the phyla to which each of the following belong

a. **Roundworms –** phylum nematode
b. **Snails** – phylum Mollusca
c. **Paramecium** – phylum porifera
d. **Mosquitoes** – arthropods (class sporozoa)

147. List any three orders of class of animals under class amphibian.

- Apoda (without legs)
- Caudata (with tail)
- Anura (without a tail)
- Sirens (without legs)

148. Describe the four characteristics of animals under phylum chordata

- Triploblastic animals
- They are acoelomates.
- They are bilateral.
- They produce sexually.

149. Discuss how cnidarians obtain their nutritional requirements.

- Cnidarians have tentacles which contain nematocysts used to attack prey or gain food.

150. Discuss how the sponges (porifera) obtain their nutritional requirement.

- Sponges use the pores which are found around its body. Water flowing in the body, aided by the flagella, helps the pores to catch food.

151. Differentiate between asconoids and sconoids based on their general structure.

- Asconoids (sponges) don't have organs and true cell tissues while sconoids have organs and true cell tissues.

152. Name the phyla to which each belong

a. Cestoda – phylum nematode
b. Anthozoa – phylum cnidarians
c. Gastropoda – phylum molluska

153. **How do marsupials, Monotremes and placenta mammals differ in terms of embryo nourishment**.

a. Marsupials – by pouching, where an embryo is nourished by the fluid and yolk present.
b. Monotremes – they lay eggs. And the embryo feeds on the yolk present.
c. Placenta mammals – by diffusion i.e. water and other nourishments diffuse across the placenta.

154. **Dentition is one of the distinguishing characters between mammals and reptiles. Explain the significance of the different dentitions in relation to evolutionary relatedness of mammalians and reptiles based on the vertebrae family tree**.

- Different arrangement of teeth, helps in processing food in animals.
- Carnivores have sharp teeth for tearing fresh.
- Herbivores have broad teeth for gliding tough plant material.
- Omnivores have all types of teeth (incisors, canines, molars and premolars) necessary for tearing fresh and gliding tough plant material.

155. How many segments do each of the following has?

a. The 28 legged diplopod

If 2 pairs of legs = 1 segment

14 pairs = x segments

x = 7 segments

b. The 26 legged centipede

If 2 legs = 1 segment

26 legs = x segments

x = 13 segments

156. State three places of mammalian glands in mammals.

a. Neck
b. Groins
c. In the brain

157. State the three classes of triploblastic animals.

- Class Mammalia
- Class Tretoda
- Class Cestoda

158. Fill the gaps with the appropriate words according to zoology.

a. The process of fertilizing eggs in fish is called ***twitting.***

b. The sponges use ***the pores which are found outside the body*** to eject water after filter feeding.
c. Hermaphrodites are also called ***bi-sexual organisms.***

159. What are the two shapes of cinidaria?

- Polyp
- Medusa

160. Give two classes of myriapods and their characteristics.

a) Chilopoda

- Use their many legs to run from enemies.
- They have one pair of leg on each segement
- They are predators.

b) Diplopoda

- It rolls their bodies when they sense danger approaching.
- They have two pairs of legs on each body parts
- They feed on plants.

"NO GENIUS IS BORN BUT CREATED THROUGH HARD WORK."

"KNOWLEDGE WITHOUT APPLICATION IS USELESS."

"CHAMPIONS ARE NOT MADE IN GYMS. CHAMPIONS ARE MADE FROM SOMETHING THEY HAVE INSIDE THEM – A DESIRE, A DREAM, A VISION."